# CSA15 Cloud Computing and Big Data Analytics

## Local IaaS Lab: Virtual Disk Allocation and VM Hardware Compatibility

| | | | |
|---|---|---|---|
| **Course Code** | CSA15 | **Course Title** | Cloud Computing and Big Data Analytics |
| **Guided Project** | GP-VM-02 | **Submission** | Individual evidence through GitHub and Google Classroom |
| **Faculty** | Dr C. Rajesh Babu | **Employee ID** | SSETSCS415 |
| **Target Learner** | Beginner CSE student | **Estimated Duration** | 2 to 3 lab hours |
| **Primary Tool** | Oracle VirtualBox or VMware Workstation | **Cloud Link** | Connects VM hardware planning with IaaS resource configguration |

**Guided Project Pattern**
Read the idea, perform the action, observe the result, record evidence, explain what happened, extend the setup to your capstone context, and submit the finnal proof through GitHub and Google Classroom.

## Learning Outcomes

- Inspect the existing VM confinguration created in the previous guided project.
- Create or plan an additional virtual hard disk using VirtualBox or VMware.
- Record disk controller, disk format, allocated size and storage-allocation method.
- Change or document VM hardware compatibility through clone/create-new-VM planning where supported.
- Explain how virtual CPU, RAM and storage decisions affeect capstone infrastructure planning.

## Before You Start

- Completed VM from GP-VM-01 using VirtualBox or VMware.
- At least 3 GB free host disk space for an additional test disk, or a writteen storage-aware plan if the laptop is constrained.
- Guest OS boot proof and hypervisor setteings screenshots from the previous guide.
- GitHub account and Google Classroom access.

## Quick Learning Notes

- A virtual hard disk is a finle on the host system that behaves like a physical disk inside the VM.
- Dynamically allocated disks grow as data is written; finxed-size disks reserve storage earlier and may perform more predictably.
- A disk controller such as SATA, SCSI or NVMe definnes how the guest OS communicates with the virtual disk.
- Hardware compatibility controls which virtual hardware version/features the VM can use. In some tools it is changed through clone, export, import or new-VM creation.

## Lab Exercise Coverage

| Lab Exercise | How This Guided Project Covers It |
|---|---|
| **Experiment 9** | Create a virtual hard disk and allocate storage using the selected hypervisor. |
| **Experiment 12** | Change or document hardware compatibility using clone/create-new-VM strategy for an already configgured VM. |

## Hypervisor Choice

| Tool | When to Choose It | Student Evidence |
|---|---|---|
| **Oracle VirtualBox** | Choose when the existing VM was created in VirtualBox. | Storage settiings, Controller, Add Hard Disk, VDI/size and VM settiings evidence. |
| **VMware Workstation** | Choose when the existing VM was created in VMware. | VM Settiings, Add Hardware, Hard Disk, SCSI/SATA/NVMe choice, VMDK/size evidence. |

**Safety Rule**
Do not atteach a host physical disk. Add only a new virtual disk finle for this lab. Avoid expanding disks beyond available host storage.

## Architecture View

| Existing VM | Settings Panel | Virtual Controller | New Virtual Disk | Compatibility Choice | Evidence |
|---|---|---|---|---|---|

Thiis lab focuses on how virtual hardware is modifined afteer a VM already exists. Thie goal is to understand storage allocation and compatibility decisions without risking the host system.

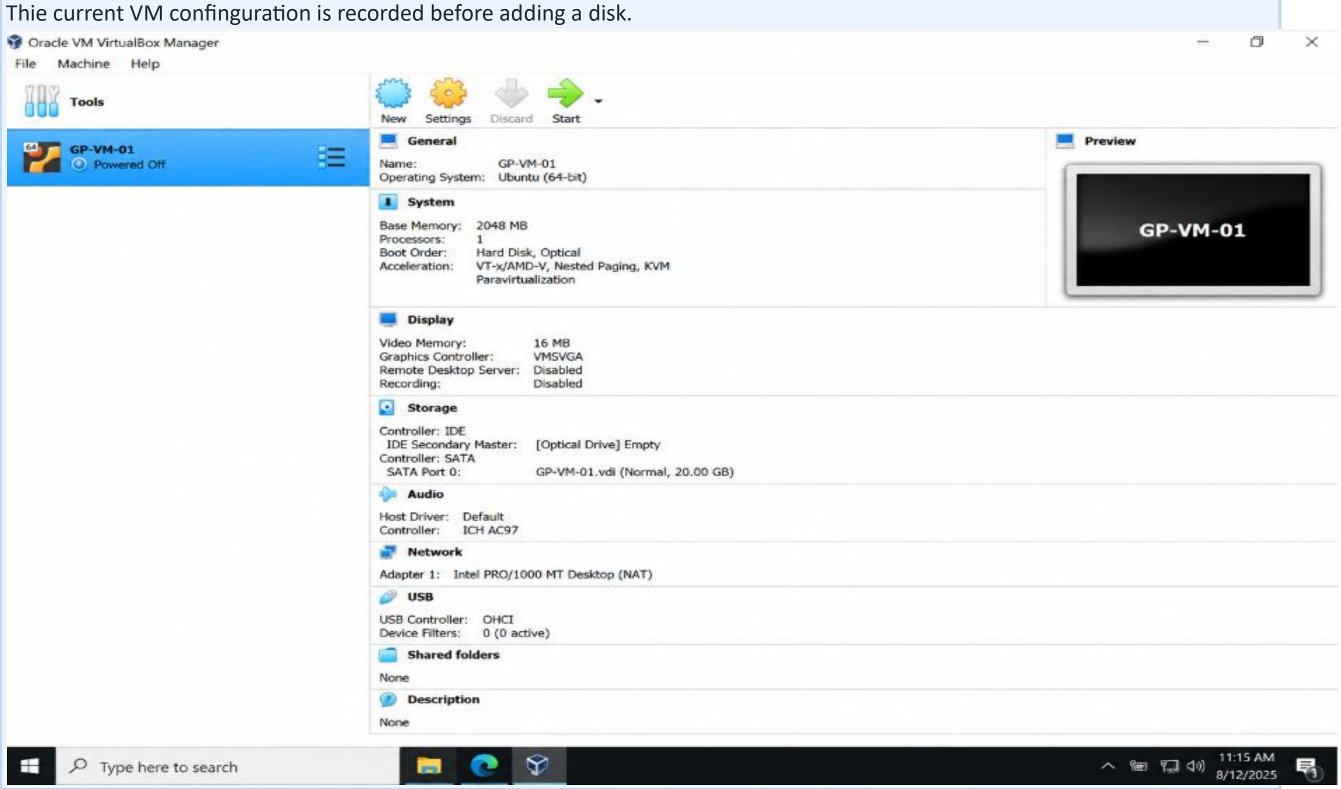
## Guided Build

### Stage 1 - Inspect the Existing VM

**Learn: Before changing hardware, the current VM state must be recorded.**

- Open VirtualBox or VMware Workstation.
- Select the VM created in GP-VM-01.
- Record VM name, selected OS, CPU, RAM, current disk size, disk format and network mode.
- Shut down the VM before changing storage hardware.

**Checkpoint**
Thie current VM confinguration is recorded before adding a disk.

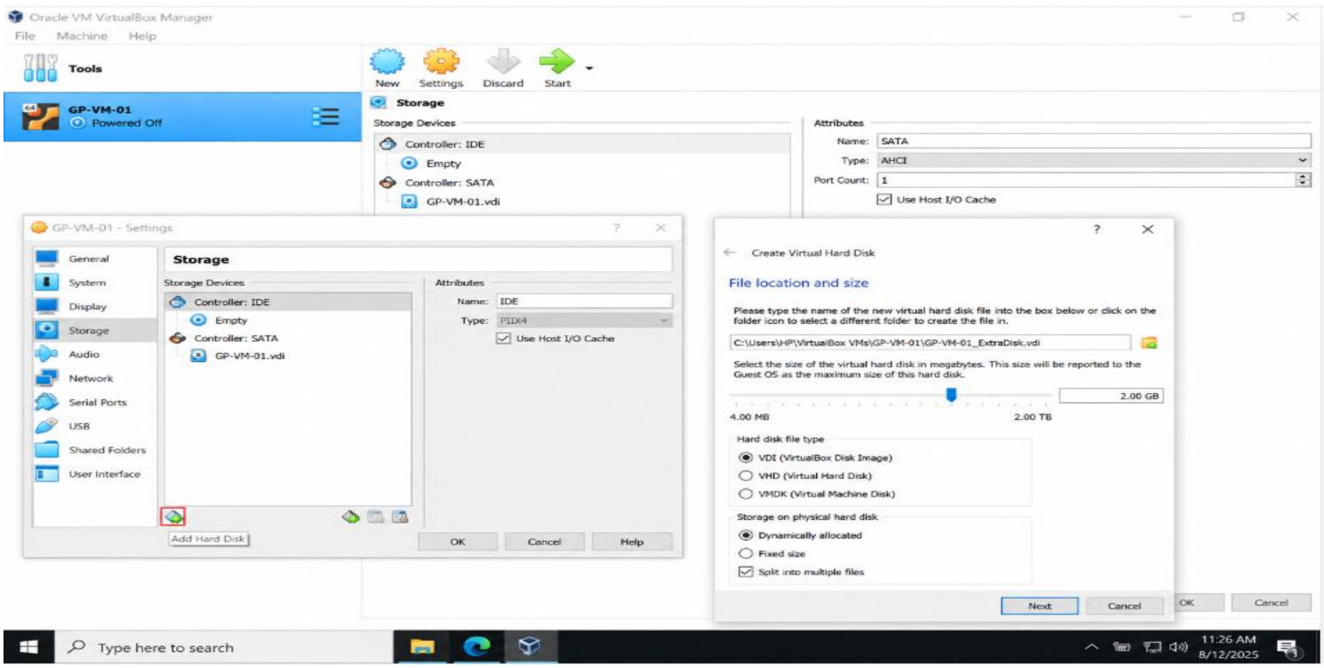


## Stage 2 - Open the Storage or Hardware Wizard

**Learn: Virtual storage is added from the hypervisor settings, not from the guest OS installer.**

- In VirtualBox, open Setteings -> Storage, select the controller area and choose Add Hard Disk.
- In VMware, open VM Setteings -> Add Hardware -> Hard Disk.
- Choose Create New Virtual Disk.
- Select a safe test size such as 2 GB unless the faculty asks for another value.
- Choose split disk into multiple finles if storage portability is preferred, or a single finle if the tool/lab requires it.

**Checkpoint**
Thie disk creation screen shows a new virtual disk size and format before finnal confinrmation.



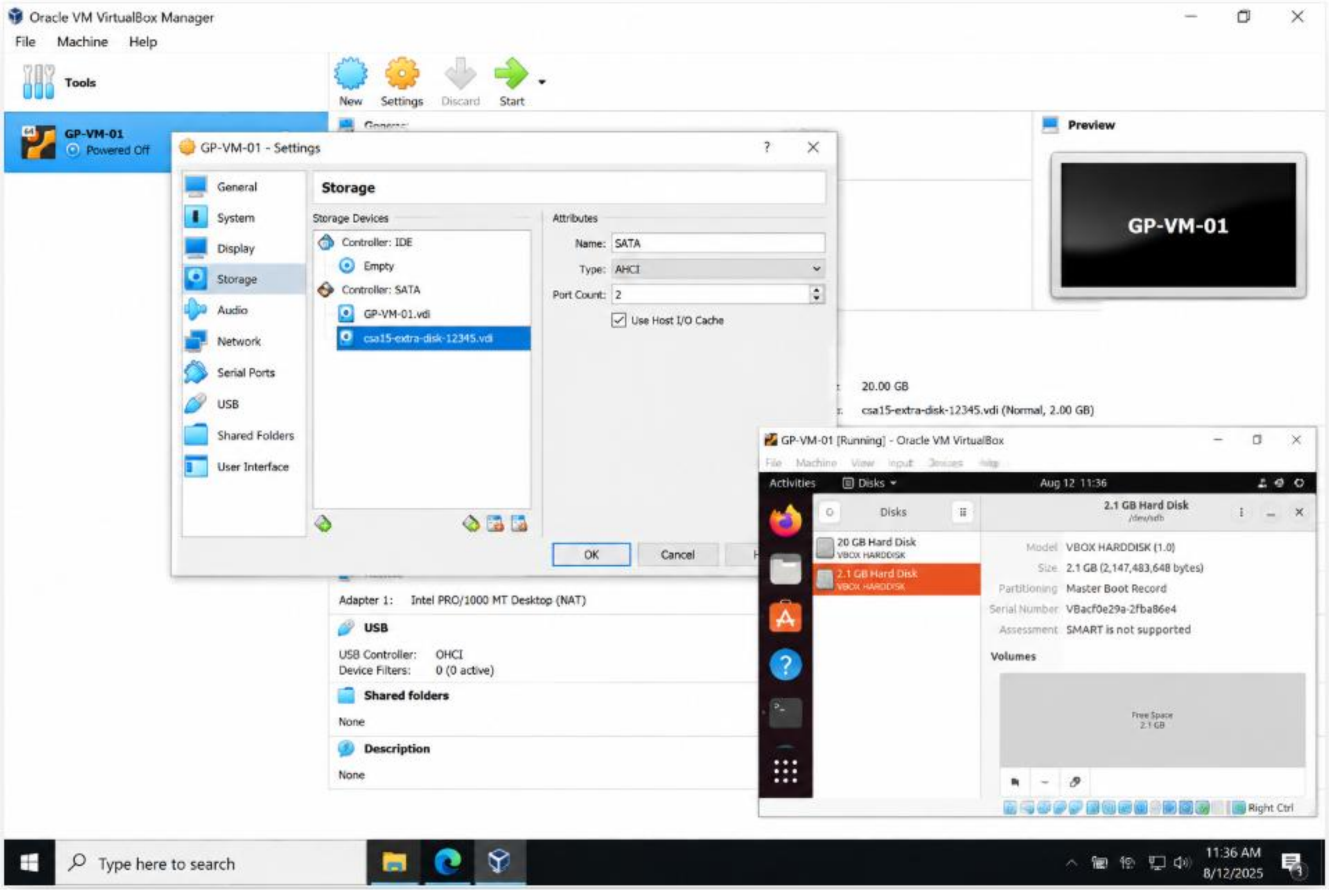

## Stage 3 - Allocate and Attach the Virtual Disk

**Learn: A newly created virtual disk must be attached to the VM through a controller before the guest OS can use it.**

- Confinrm the new disk finle name using register number, for example csa15-extra-disk-register-number.
- Atteach the disk to a suitable controller: SATA is acceptable in VirtualBox; SCSI/SATA/NVMe may appear in VMware depending on compatibility.
- Save setteings and capture the storage/setteings page showing both original disk and new disk.
- Boot the VM and observe whether the guest OS detects an additional disk. Formatteing inside the guest OS is optional and should be done only if instructed.

**Checkpoint**
Thie hypervisor setteings show the additional virtual disk atteached to the existing VM.

## Record 3 - Attached virtual disk evidence



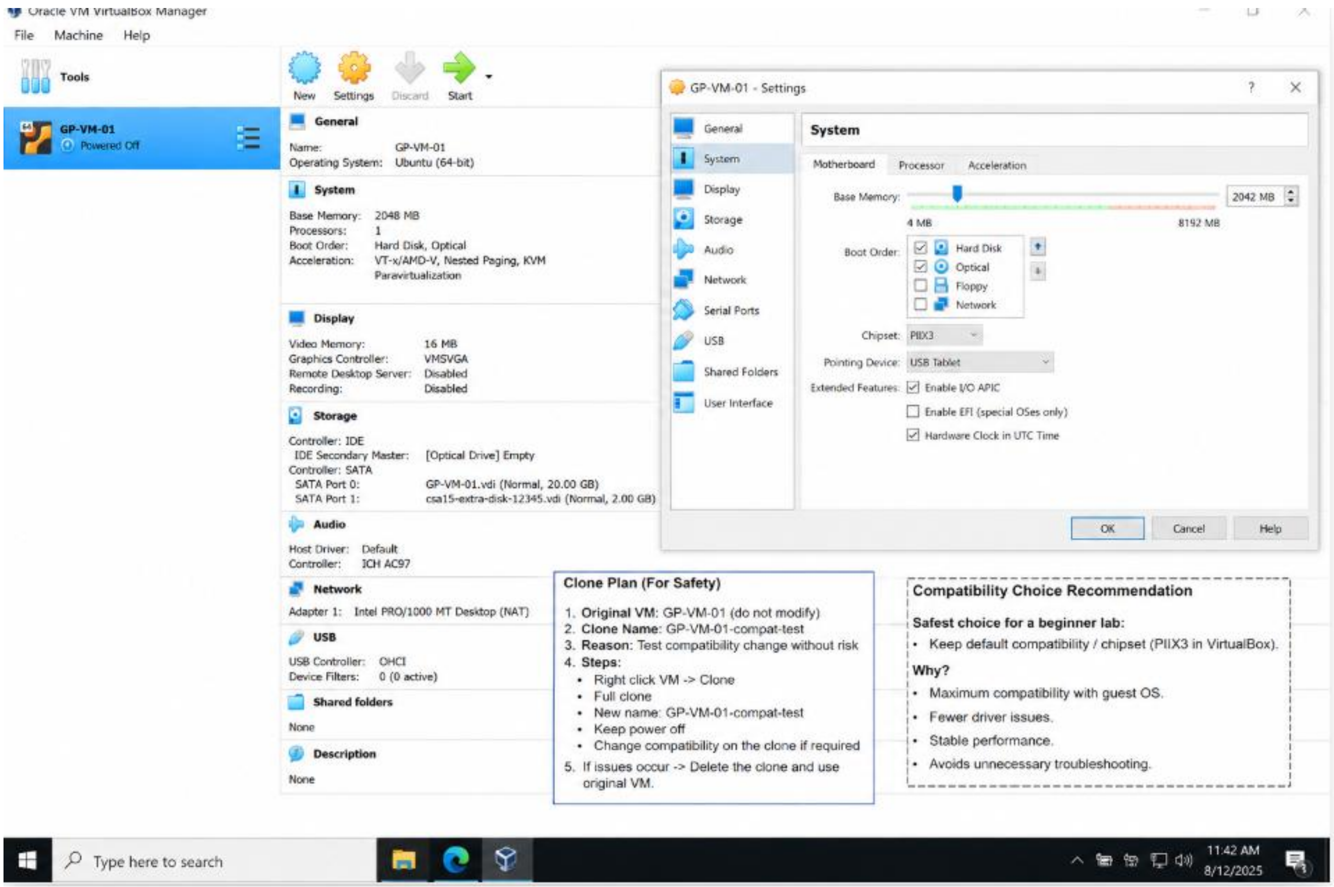

## Stage 4 - Hardware Compatibility Review

**Learn: Hardware compatibility controls which virtual hardware features and controller types are available.**

- In VMware, inspect VM -> Manage -> Change Hardware Compatibility if available. If it is restricted, record the menu or version limitation.
- In VirtualBox, record equivalent compatibility-related setteings such as System, Acceleration, Storage Controller and chipset options.
- If the tool recommends cloning before compatibility change, create a clone plan rather than modifying the only working VM.
- Write which compatibility choice is safest for a beginner lab and why.

**Checkpoint**
Hardware compatibility evidence or a justified compatibility-change plan is documented.

## Record 4 - Hardware compatibility evidence



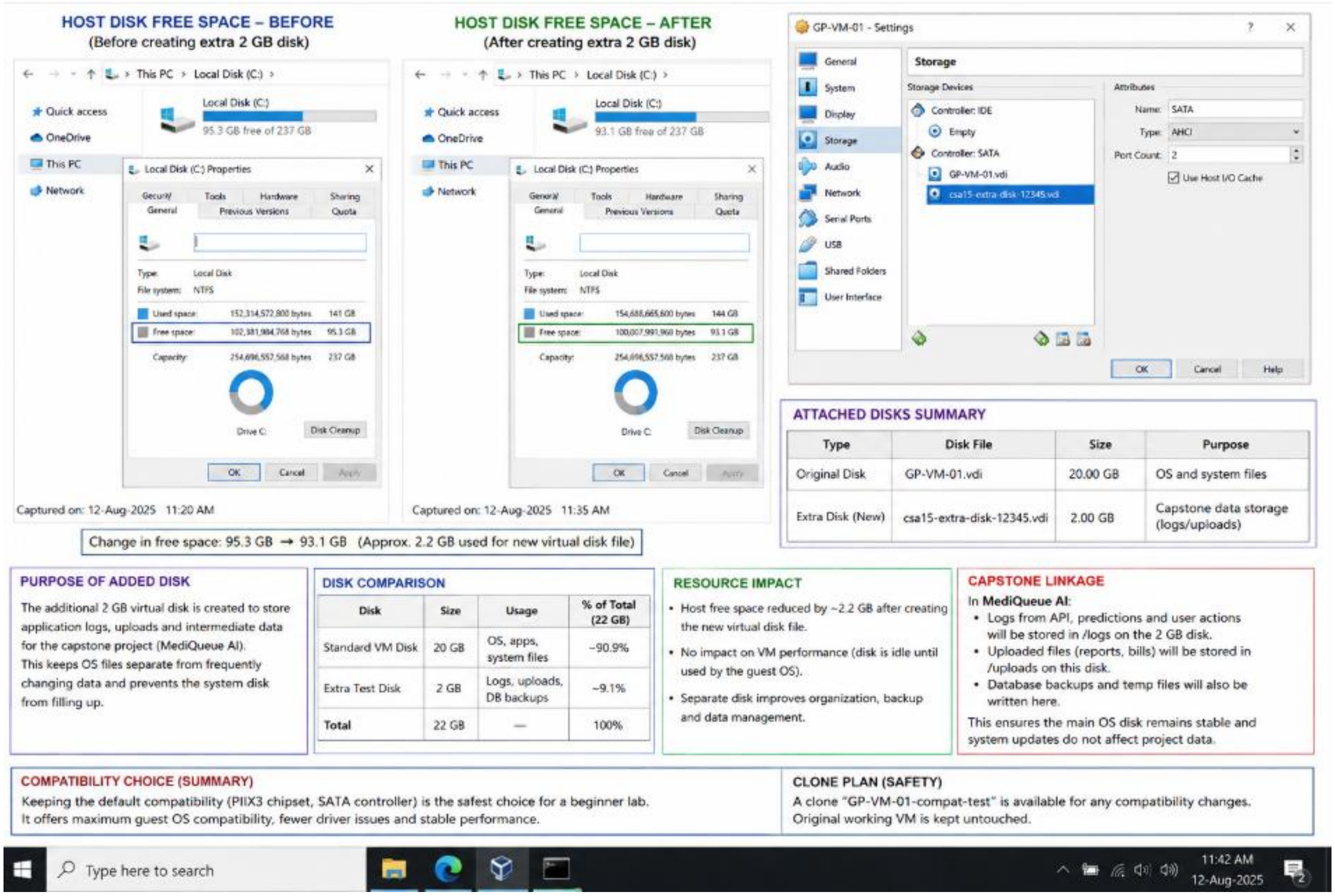

## Stage 5 - Resource Impact and Capstone Link

**Learn: Extra disk and hardware choices must be justified by workload, not added randomly.**

- Record host free disk space before and afteer creating the additional disk if possible.
- Write the purpose of the added disk: logs, uploads, database finles, backup, test data or capstone module storage.
- Compare standard 15 GB VM disk with the extra 2 GB test disk.
- Connect this storage decision to one capstone data requirement.

| Checkpoint |
| --- |
| Thie student explains why the additional disk or compatibility decision is useful. |

## Record 5 - Resource impact and capstone linkage




## Final Evidence Checklist

| Done | Checkpoint | Evidence to Capture |
|---|---|---|
| ☐ | **Baseline captured** | Existing VM settiings before modificcation. |
| ☐ | **Additional disk planned/created** | Virtual disk wizard screenshot with size and format. |
| ☐ | **Disk attached** | Settiings page showing original disk and additional disk. |
| ☐ | **Compatibility reviewed** | VMware compatibility menu or VirtualBox equivalent settings evidence. |
| ☐ | **Capstone link written** | Storage purpose and resource-impact explanation. |

# Explain and Extend

## Explain the difference between a virtual hard disk file and a physical disk.

| Virtual Hard Disk | Physical Disk |
|---|---|
| A **file** created by software such as VirtualBox or VMware. | A **real hardware storage device** such as an HDD or SSD. |
| Used by a **virtual machine** as its storage. | Used directly by the host operating system or computer. |
| Its size can be dynamically allocated or fixed. | Has a fixed physical capacity. |

| Virtual Hard Disk | Physical Disk |
| --- | --- |
| Easy to copy, move, clone, or delete. | Cannot be copied like a normal file; data must be cloned or transferred. |
| Example: csa15-extra-disk-12345.vdi – **2 GB**. | Example: a **256 GB SSD** installed in the computer. |

## Explain why dynamically allocated storage is useful for beginner labs.

Dynamically allocated storage means the virtual disk is given a maximum size, but the physical storage on the host computer is used only as the VM actually needs it.

For example, if you create a 20 GB virtual disk, it may initially occupy only a few GB on the host. As you add files inside the VM, the virtual disk file grows until it reaches 20 GB.

### Advantages

1. Saves host storage – It does not immediately consume the entire maximum disk size.
2. Easy to create – Beginners can create a disk without needing large free space immediately.
3. Flexible – The disk grows automatically as more data is added.
4. Good for experiments – Useful when testing different OS configurations and applications.
5. Less resource waste – Unused virtual disk space does not occupy the host unnecessarily.

### Example

If a VM has a 20 GB dynamically allocated disk:

Maximum size: 20 GB
Actual data used: 5 GB
Host storage consumed: approximately 5 GB

So, for beginner labs, dynamically allocated storage is a simple, flexible, and space-efficient choice.

## Explain why compatibility changes should be tested on a clone or documented plan.

Compatibility changes should be tested on a clone because they may cause boot or hardware problems. A clone protects the original working VM and allows safe testing.

Example: Keep GP-VM-01 unchanged and test changes on GP-VM-01-compat-test.

## Extend this setup to your capstone: what data could be stored on a separate disk?

A separate disk can store **logs, uploaded files, databases, backup files, test data, and application files**. This keeps data organized and separate from the operating system.

## Submission Protocol

- Create a GitHub folder named CSA15-GP-VM02-Virtual-Disk-Hardware-Compatibility.
- Upload this completed document as DOCX/PDF along with screenshots and short README.md.
- README.md must include tool used, VM/cloud specifincations, confinguration decisions, screenshots index and cleanup status.
- Submit the GitHub repository link in Google Classroom. Do not upload passwords, private keys, tokens, public IP credentials or sensitive account screenshots.

## Student Assurance

**Declaration**

I confirm that this guided-project evidence was completed by me, the screenshots are from my own lab environment, and no secret keys, passwords or sensitive account information are included.

Student Signature: **T.PRAKEERTHI** Register Number: 192524279 Date: 14/08/2026